

Buffer Stock Saving with Heterogeneous Agents

An example of the Econ-ARK MyST template

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For everyone who waited for the solver to converge.

All models are wrong, but some are useful. George Box

Abstract We solve a buffer stock saving model with the endogenous grid method and show how the Econ-ARK template typesets a working paper: title block, margin notes, equations, tables, code and author-year citations.

Summary Households save more when their income is uncertain. This paper shows how much more, and how quickly the extra saving can be computed.

Keywords consumption, precautionary saving, heterogeneous agents

JEL codes C63, D14, E21

Materials

Run online
Launch
Starts in a few minutes

Code
econ-ark/econ-ark-
myst
MIT license

REMARK
BufferStockTheory

Source
github.com/econ-
ark/econ-ark-myst

Also as
Latest version

1 Introduction

Precautionary saving arises when income risk interacts with a convex marginal utility (Carroll 1997). The endogenous grid method (Carroll 2006) makes such models cheap to solve, because it replaces root finding with an inversion of the Euler equation.¹ Households hold a buffer of wealth against bad income draws. The size of that buffer depends on patience, the return on saving and the degree of risk.

2 Model

The consumer maximizes expected discounted utility subject to the budget constraint

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¹The method inverts the first-order condition on a grid of end-of-period assets rather than solving it on a grid of market resources. `IndShockConsumerType` applies it to Section 2's model.

Key points

- A buffer of wealth protects households against bad income draws.
- The endogenous grid method solves the model without root finding.
- Patience, returns and risk set the size of the buffer.

Working paper

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$$v(m_t) = \max_{c_t} u(c_t) + \beta \mathbb{E}_t[v(m_{t+1})], \quad m_{t+1} = R(m_t - c_t) + y_{t+1}. \quad (1)$$

The Euler equation implied by (1) is $u'(c_t) = \beta R \mathbb{E}_t[u'(c_{t+1})]$, and with constant relative risk aversion $u(c) = c^{1-\rho}/(1-\rho)$.

2.1 Calibration

Table 1 Baseline calibration.

Parameter	Description	Value
β	Discount factor	0.96
R	Gross interest factor	1.03
ρ	Relative risk aversion	2.00

2.1.1 Sources of the parameters

The discount factor and the interest factor follow [Carroll \(1997\)](#), and risk aversion is set to the value the literature uses most often.

3 Results

Table 1 lists the parameters used in the solution, which come from the 1990's literature. Proposition 1 describes the shape of the solution.

Proposition 1 (Concavity). *If income risk is present, the consumption function is strictly concave. Its slope falls toward the perfect foresight marginal propensity to consume as wealth grows.*

Proof. See [Carroll \(1997\)](#) for the argument under constant relative risk aversion. $\square$

Note

Admonitions take a rule in the palette rather than the filled box MyST draws by default.

```
from HARK.ConsumptionSaving.ConsIndShockModel import
IndShockConsumerType

agent = IndShockConsumerType()
agent.solve()
```

Appendix A derives the Euler equation.

Declarations

The authors report no competing interests. The authors used a large language model to draft the example text of this template and reviewed every sentence before release.

Acknowledgments

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Data availability

All code needed to reproduce the results is in the repository linked in the margin.

References

- Carroll, Christopher D. 1997. "Buffer-Stock Saving and the Life Cycle/Permanent Income Hypothesis." *The Quarterly Journal of Economics* 112 (1): 1–56. <https://doi.org/10.1162/003355397555109>.
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Appendix A Derivation of the Euler equation

The first-order condition for c_t in (1) sets $u'(c_t)$ equal to $\beta R E_t[v'(m_{t+1})]$. The envelope condition gives $v'(m_t) = u'(c_t)$, and substituting it yields the Euler equation.